Engine Model(s): PT6A

73-10-04

FLOW DIVIDER AND STARTING CONTROL - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC01-001	Fuel, Engine
PWC05-061	Cloth, Abrasive Coated
PWC05-101	Cloth, Abrasive
PWC05-166	Solution, Chemical Treatment
PWC05-168	Compound, Polishing
PWC05-344	Cable, Safety, Kit
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine
PWC11-038	Solvent, Cleaning

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC90025	Crimper, Safety Cable

4. Fixtures, Equipment and Supplier Tools

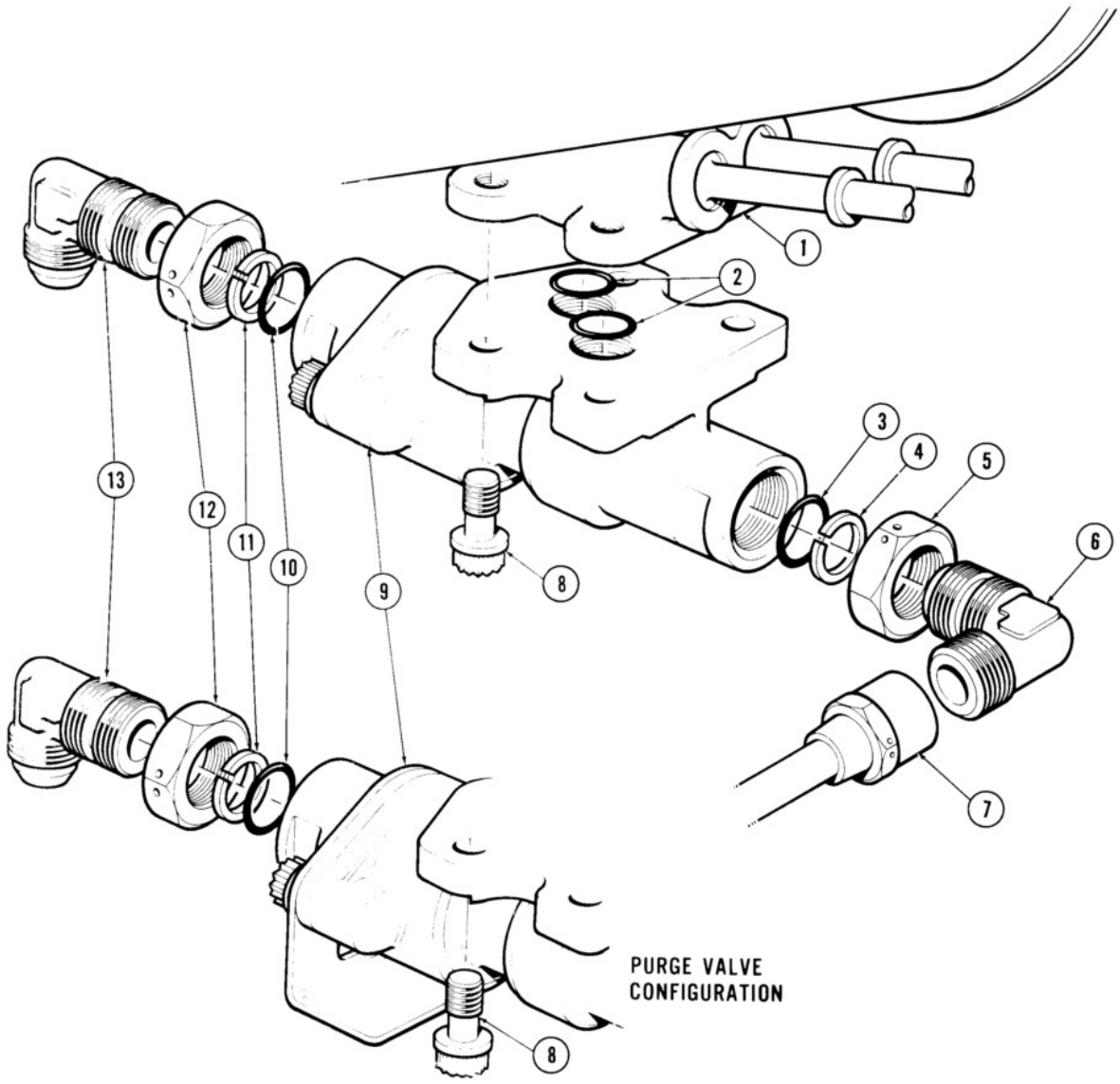
Not Applicable

5. Removal/Installation

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A. Removal of Flow Divider and Dump (Pre-SB1406) or Purge (Post-SB1406) Valve (PT6A-21 Engines) (Ref. Fig. 201)

Figure 201 Removal/Installation of Flow Divider and Dump (Pre-SB1406) or Purge (Post-SB1406) Valve



PT6A-21

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1. Inlet Adapter
2. Preformed Packing
3. Preformed Packing
4. Back-up Ring
5. Jam Nut
6. Elbow
7. Fuel Inlet Line
8. Bolt
9. Flow Divider and Dump or Purge Valve
10. Preformed Packing
11. Back-up Ring
12. Jam Nut
13. Elbow

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NOTE: 1. To avoid contamination of work area, use a container to catchfluid when disconnecting fuel lines, and during removal of flow divider and purge valve.

NOTE: 2. Smoky starts may be alleviated by replacing existing flow divider and purge valve with one having a larger diameter hole in the outlet fitting (P/N 3035229, 3036641, Ref. IPC).

(1) If flow divider and dump/purge valve (9) is being removed with engine installed, disconnect dump line or purge air line from elbow (13) on right side of engine. Cap line connection and elbow.

(2) Disconnect coupling nut of fuel inlet tube (7) at elbow (6) on left side of engine. Cap line connection and elbow.

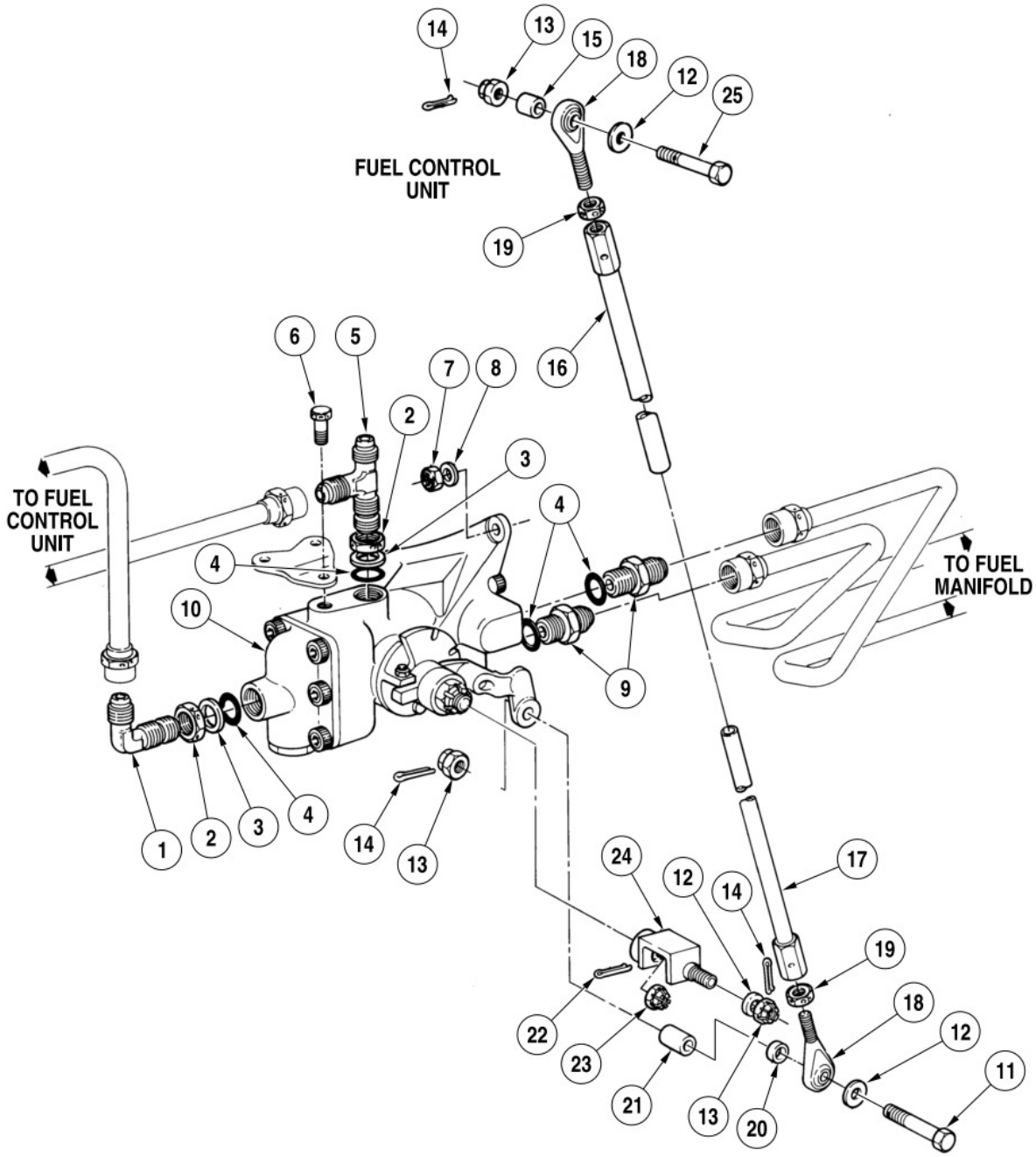
(3) Support valve (9) and remove four bolts (8) that secure valve to fuel manifold inlet adapter (1). Remove valve and discard preformed packings (2).

(4) If flow divider and dump/purge valve is being replaced, note angular position of elbows (6 and 13), then remove elbow fittings; discard preformed packings (3 and 10).

B. Removal of Starting Flow Control (PT6A-27 and PT6A-28 Engines) (Ref. Fig. 202)

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Figure 202 Removal/Installation of Starting Flow Control



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1. Elbow
2. Nut
3. Back-up Ring
4. Preformed Packing
5. Tee
6. Bolt
7. Nut
8. Washer
9. Straight Nipple
10. Starting Flow Control
11. Bolt
12. Washer
13. Castellated Nut
14. Cotterpin
15. Spacer
16. Upper Rod Assembly
17. Lower Rod Assembly
18. Rod End Connector
19. Nut
20. Sleeve Spacer
21. Sleeve Spacer
22. Cotterpin
23. Castellated Nut
24. Extension
25. Bolt

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NOTE: To prevent contamination of work area, use a container to catch spillage when disconnecting fuel lines and during removal of flow control.

(1) Remove cotterpin (14), castellated nut (13), washer (12), spacers (20 and 21) and bolt (11) that secure lower end of starting flow control rod assembly to starting flow control (10).

(2) Remove cotterpin (14), castellated nut (13), washer (12) and spacer (15) that secure upper end of starting flow control rod assembly to fuel control unit. Remove rod assembly.

(3) Disconnect coupling nuts of fuel pressure and bypass tubes from elbow (1) and tee adapter (5) respectively. Cap all tubes and openings.

(4) Disconnect coupling nuts of primary and secondary fuel tubes from nipples (9) on starting flow control. Cap all tubes and openings.

(5) Remove two self-locking nuts (7) and washers (8) that secure starting flow control to studs on flange G of engine, and remove single bolt (6) securing starting flow control to bracket attached to engine.

(6) If starting flow control is to be replaced, remove straight nipples (9), elbow (1), tee-adapter (5) and bracket (24) from unit; discard preformed packings. Retain all other hardware for installation on replacement unit.

C. Installation of Flow Divider and Dump (Pre-SB1406) or Purge (Post-SB1406) Valve (PT6A-21 Engines) (Ref. Fig. 201)

(1) When a replacement flow divider and dump/purge valve (9) is being installed, depreserve unit and assemble elbows (6 and 13) prior to installation:

NOTE: Depreservation is not required when original flow divider and dump/purge valve is being installed.

(a) Remove shipping plugs from inlet and dump line or purge air ports of replacement valve, and drain as much preservation oil as possible from unit. Make sure that preservation oil is present if the unit is a replacement flow divider and dump or purge valve. If preservation oil is not present in the replacement flow divider and dump or purge valve, the unit must be returned to a P&WC approved overhaul facility for inspection.

(b) Using fuel (PWC01-001), flush out valve from dump line or purge air port end to remove all residual preservation oil from this area of the valve. Continue flushing until clear fuel flows from primary and secondary manifold ports. Temporarily reinstall plug in dump or air port.

(c) Flush out valve at area of fuel inlet port with fuel, and ensure all residual preservation oil is removed. Temporarily reinstall plug in inlet port.

(d) Assemble nuts (5 and 12), and back-up rings (4 and 11) followed by preformed packings (3 and 10) on each elbow (6 and 13) (Ref. 70-00-00, REMOVAL/INSTALLATION).

(e) Remove plugs from inlet and dump or purge air ports, install elbows (6 and 13) in valve (9), and position each elbow at same angle noted during removal. Tighten nuts (5 and 12), torque 38 to 42 lb.in. and safety wire.

(f) To prevent ingress of foreign material, cap elbows (6 and 13) until valve (9) is ready for

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installation.

(2) Press two preformed packings (2) into recesses at primary and secondary manifold ports in mounting face of valve (9).

(3) Position valve (9) on base of fuel manifold inlet adapter (1), align attachment holes in each part, and secure valve to adapter with four bolts (8). Tighten bolts, torque 32 to 36 lb.in. and safety wire bolts together in pairs.

(4) Remove blanking caps from fuel pressure tube (7) and elbow (6), and connect coupling nut to elbow. Tighten coupling nut, torque 90 to 100 lb.in. and install safety wire.

(5) If installation of flow divider and dump/purge valve has been carried out with engine installed, remove blanking caps from dump line or purge air line and from elbow (13) and connect line to elbow. Tighten line connection to torque value recommended in relevant Aircraft Maintenance Manual.

D. Installation of Starting Flow Control (PT6A-27 and PT6A-28 Engines) (Ref. Fig. 202)

(1) When a replacement starting flow control (10) is being installed, depreserve unit and assemble elbows, tee and nipples prior to installation:

NOTE: Depreservation is not required when original starting flow control (10) is being installed.

(a) Remove shipping plugs from inlet and dump ports of replacement unit and drain as much preservation oil as possible from unit.

(b) Using fuel (PWC01-001), flush out unit from dump port end to remove all residual preservation oil from this area of the unit. Continue flushing until clear fuel flows from primary and secondary manifold ports. Temporarily reinstall plug in dump port.

(c) Flush out valve at area of inlet port with fuel and ensure all residual preservation oil is removed. Temporarily reinstall plug in inlet port.

(d) Assemble nuts (2) and back-up rings (3) followed by preformed packings (4) on elbow (1) and tee (5) fittings.

CAUTION: WHEN YOU TORQUE THE T-FITTING JAM NUT, APPLY AN EQUAL TORQUE ON THE T-FITTING IN THE OPPOSITE DIRECTION. THIS WILL PREVENT THE DEVIATION OF THE T-FITTING INSTALLATION FROM THE RECORDED ANGLE. A DEVIATION IN POSITION CAN CAUSE DAMAGE TO THE AIRCRAFT RIGID FUEL DUMP LINE.

(e) Remove the plugs from the inlet and dump ports, install the elbow (1) and tee (5) in the starting flow control (10), and position each fitting at the same angle noted during removal. Torque the two jam nuts (2) 38 to 42 lbf.in.

(f) Remove the plugs from the primary and secondary fuel ports and install the straight nipples (9) and preformed packings (4) in the ports. Torque the nipples 65 to 75 lbf.in.

(g) To prevent ingress of foreign material, cap elbows, tee and nipples until control (10) is ready for installation.

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(2) Install the starting flow control (10) on the studs at Flange G of the accessory gearbox:

(a) Remove the caps from the elbow (1), tee (5) and nipples (9).

(b) Attach the starting flow control (10) with the two self-locking nuts (7) and washers (8).

(c) Torque the nuts 36 to 40 lbf.in.

WARNING: PUT ON THE EYE PROTECTION WHEN YOU INSTALL THE SAFETY CABLE. PIECES CAN BREAK OFF AND CAUSE INJURY.

(3) Attach the bracket on the accessory gearbox to the upper face of the starting flow control (10) with the single bolt (6). Torque the bolt 32 to 36 lbf.in. and lock with the safety cable kit (PWC05-344) and the crimper (PWC90025).

WARNING: PUT ON THE EYE PROTECTION WHEN YOU INSTALL THE SAFETY CABLE. PIECES CAN BREAK OFF AND CAUSE INJURY.

(4) Connect the coupling nuts of the primary and secondary fuel tubes to the respective nipples (9) on the starting flow control (10). Torque the nuts 90 to 100 lbf.in. and lock with the safety cable kit (PWC05-344) and the crimper (PWC90025).

(5) Install the pressure and bypass fuel tubes onto the starter control unit (10):

(a) Connect the coupling nut of the pressure fuel tube to the elbow (1).

(b) Connect the coupling nut of the bypass fuel tube to the tee fitting (5).

(c) Loosen the jam nut (2) to allow the angle of the tee fitting (5) and the elbow (1) to be adjusted if required. Torque the jam nut again (Ref. step (1)).

(d) Tighten the coupling nuts of the fuel tubes with your hand.

(e) Torque the coupling nuts of the fuel tubes (Ref. 73-10-03, FUEL LINES - MAINTENANCE PRACTICES).

WARNING: PUT ON THE EYE PROTECTION WHEN YOU INSTALL THE SAFETY CABLE. PIECES CAN BREAK OFF AND CAUSE INJURY.

(f) Attach the coupling nuts of the bypass and pressure fuel tubes and the jam nuts (2) with the safety cable kit (PWC05-344) and the crimper (PWC90025).

(6) If engine is mounted in airframe, connect fuel dump line to tee fitting (5). Tighten and torque line fitting as detailed in relevant Aircraft Maintenance Manual.

(7) Install the upper end of the starting flow control rod at the fuel control unit and attach to the lever with the bolt (25), spacer (15), washer (12) and castellated nut (13). Torque the nut 12 to 18 lbf.in, do not lock with the cotterpin (14) at this stage.

(8) Install the lower end of starting flow control rod at the starting flow control (10) lever, and attach with the bolt (11), spacers (21 and 20), washer (12) and castellated nut (13). Torque the nut 12 to 18 lbf.in., do not lock with the cotterpin (14) at this stage.

(9) Adjust length of the control rod (Ref. 76-10-00), torque the castellated nuts (13) again 12 to 18 lbf.in., lock with the cotterpins (14).

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6. Cleaning/Painting

A. Cleaning of Flow Divider

(1) Clean exterior surfaces of the flow divider and dump/ purge valve or starting control with petroleum solvent (PWC11-027) or (PWC11-031). To prevent cleaning agent from entering valve, do not remove blanking caps.

(2) Wipe surfaces dry with clean, lint-free cloths to remove all residual solvent and contamination.

7. Inspection/Check

A. General Inspection

(1) Examine the mounting flange for cracks and other defects. Cracks are not permissible; minor surface damage may be repaired.

(2) Examine machined face of mounting flange for gouges, nicks, scores, scratches, and similar defects which can prevent correct mating of flange surface with fuel manifold inlet adapter. Particular attention should be paid to the two recesses for primary and secondary manifold port seals.

(3) If elbows have been removed from inlet and dump ports, examine each port for thread damage.

(4) Check for signs of leakage at joint between valve housing and main body. If leakage is evident, return valve to supplier (Ref. IPC).

(5) Examine surfaces of valve for corrosion and for general condition of anodic treatment.

(6) On the flow divider and dump or purge valve, check for signs of leakage at the spigot joint between the dump spring housing or purge valve housing and main body. If leakage is evident, return the valve to an approved overhaul facility.

(7) On the starting control, check the lever for freedom of movement. Suspect units should be returned to an approved overhaul facility.

(8) Inspect the surfaces of the valve or starting control for corrosion and for the general condition of the anodic treatment.

8. Approved Repairs

A. General Repair

NOTE: Make sure that all areas are cleaned and corrosion preventive treatment applied to surfaces after carrying out minor repairs. Do not allow filings to enter the valve.

(1) Minor damage such as burrs, nicks, scores, scratches and other similar defects, may be cleaned up by blending with a fine stone or crocus cloth (PWC05-061), making sure that all high spots and sharp edges are removed.

(2) Minor defects on the machined face of the flow divider mounting flange may be cleaned up by lightly dressing with crocus cloth (PWC05-061) held to a true, flat plate.

(3) Clean up minor thread damage in the fuel inlet and dump or air inlet ports, as applicable, or

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chase the threads 7/16-20 UNJF-3B.

(4) Clean up minor thread damage on the elbow fittings or chase the threads using the appropriate die.

B. Corrosion Removal

(1) Light surface corrosion on the units may be removed from the main body and from the dump spring or purge valve housing (as applicable) using one of the following methods:

NOTE: Apply anti-corrosion treatment to the repaired area in accordance with the procedure given in Subpara. C.

(a) Remove corrosion deposits by vapor blasting with 1200 grit (PWC05-168) or finer. Make sure that the areas around the corrosion are masked to prevent damage to the surrounding parts.

(b) Remove corrosion deposits by careful polishing with No. 400 grit abrasive cloth (PWC05-101) followed by crocus cloth (PWC05-061).

C. Application of Anodize Surface Treatment

(1) The anodic anti-corrosion finish on the unit surfaces of the main body and dump spring or purge valve housing (as applicable) may be repaired by local application of the chemical treatment solution (PWC05-166). This treatment must be used on areas where minor repairs have been carried out by local blending and/or polishing. To apply the treatment, proceed as follows:

(a) Prepare the surface by cleaning with the cleaning solution (PWC11-038). Keep the surface wetted with the cleaning solution for one to five minutes.

(b) Rinse with clean water to remove all traces of the cleaning solution.

WARNING: ALODINE SOLUTION IS EXTREMELY DANGEROUS. IT CONTAINS AN OXIDIZING INGREDIENT WHICH CAN CAUSE AN EXPLOSION IF IT COMES IN CONTACT WITH COMBUSTIBLE MATERIALS SUCH AS PAINTS AND SOLVENTS.

WARNING: DURING THE USE OF ALODINE SOLUTION, AVOID BREATHING VAPORS; USE ONLY WITH ADEQUATE VENTILATION. WEAR CHEMICAL SAFETY GOGGLES AND RUBBER GLOVES WHEN HANDLING. IN CASE OF SKIN CONTACT, WASH WITH SOAP AND WATER; IF IRRITATION APPEARS AND PERSISTS, CONSULT A PHYSICIAN (MEDICAL DOCTOR). IN CASE OF EYE CONTACT, FLUSH EXTENSIVELY WITH FRESH WATER AND CONSULT A PHYSICIAN IMMEDIATELY.

(c) Apply the chemical treatment solution to the prepared surface by brush, swab, or spray method and allow three to five minutes setting time.

(d) Rinse with clean water to remove all excess chemical treatment solution and allow the surface to air dry.

(e) Examine the coating and make sure the repaired area is completely covered. Reapply

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the treatment as necessary.

9. Adjustment/Test

Check the function of the flow divider and dump or purge valve or starting control, as applicable, during the next engine test run (Ref. 71-00-00).

A. Procedure

(1) Start engine (Ref. 71-00-00, ADJUSTMENT/TEST), and do checks associated with Installation/Replacement of Flow Divider (Ref. 71-00-00, ADJUSTMENT/TEST).

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